



Introduction

Background. Standard benchmarks such as IMAGENET-C apply corruptions *uniformly across the entire image*.

Motivation. Real-world artifacts are often *spatially localized*, selectively degrading objects while leaving the surrounding scene intact; this regime remains under-represented.

Contribution. We introduce **ImageNet-LC**, a benchmark for evaluating CV robustness to **object-aware, localized corruptions** applied to semantic RoIs identified by an object detector.

Key Finding. IMAGENET-LC reveals that *object-centric localised corruptions* can evade SOTA image classifiers while maintaining a minimal perceptual change.

Proposed Approach

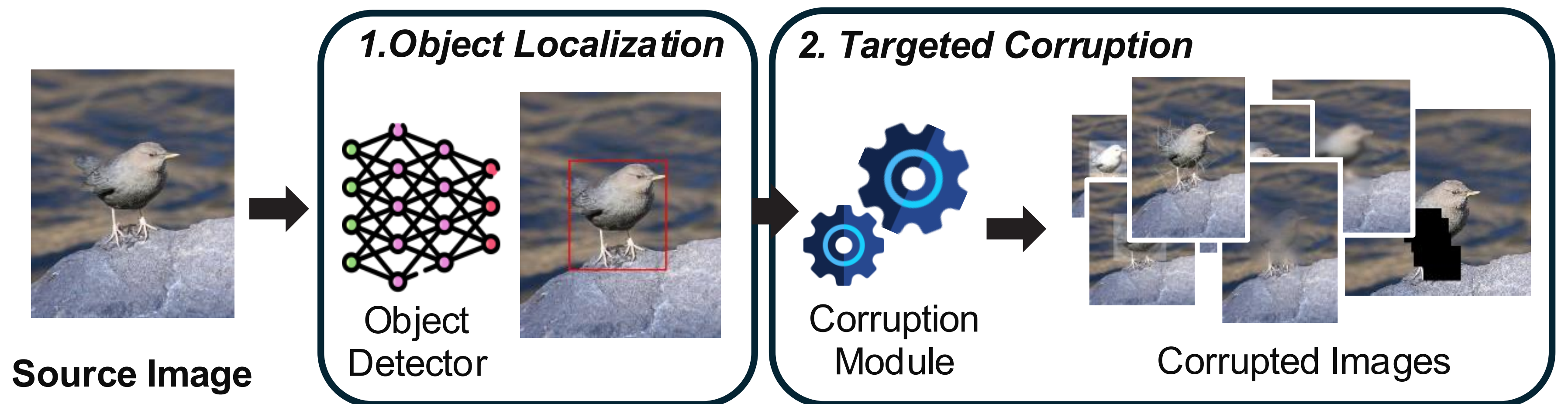


Figure 1: Overview of the object-centric corruption pipeline for IMAGENET-LC. Images are first processed by an object localization module to identify RoIs, followed by targeted application of localized corruptions within these regions.

Phase 1: Object Localization. A YOLO-based detector identifies foreground objects, with the top $k = 3$ confidence-ranked bounding boxes selected as semantic RoIs for localized corruption.

Phase 2: Targeted Corruption. Corruptions are applied independently inside each box, while the rest of the image is left untouched.

Corruption Gallery

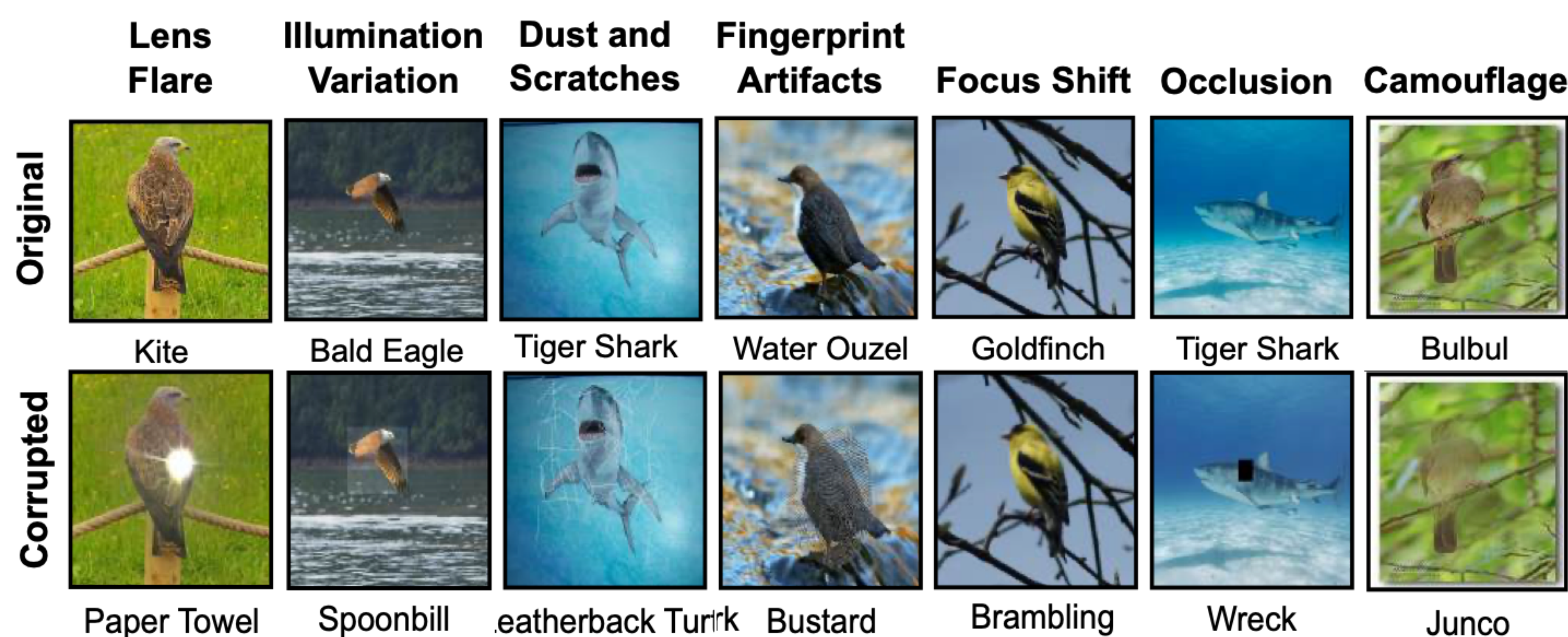


Figure 5: The seven localized corruptions applied to different objects. The label under each image is the model prediction: the object is clearly recognisable to a human but the classifier has failed.

Metrics & Setup

CE, mCE, and Relative mCE: To quantify model robustness under localized corruptions, for a model f with clean top-1 error E_{clean}^f and error $E_{s,c}^f$ at severity s under corruption c :

$$\text{CE}_c^f = \left(\sum_{s=1}^5 E_{s,c}^f \right) / \left(\sum_{s=1}^5 E_{s,c}^{\text{AlexNet}} \right)$$

mCE averages CE over the seven corruption types to provide an overall robustness measure; **Relative mCE** first subtracts each model's clean error, accounting for differences in baseline performance. **Lower is more robust.**

Perceptual Similarity: We measure perceptual similarity between original and corrupted images using LPIPS, where 0 indicates identical images and 1 indicates high perceptual difference, capturing the overall visual change caused by corruption.

Evaluation: Robustness

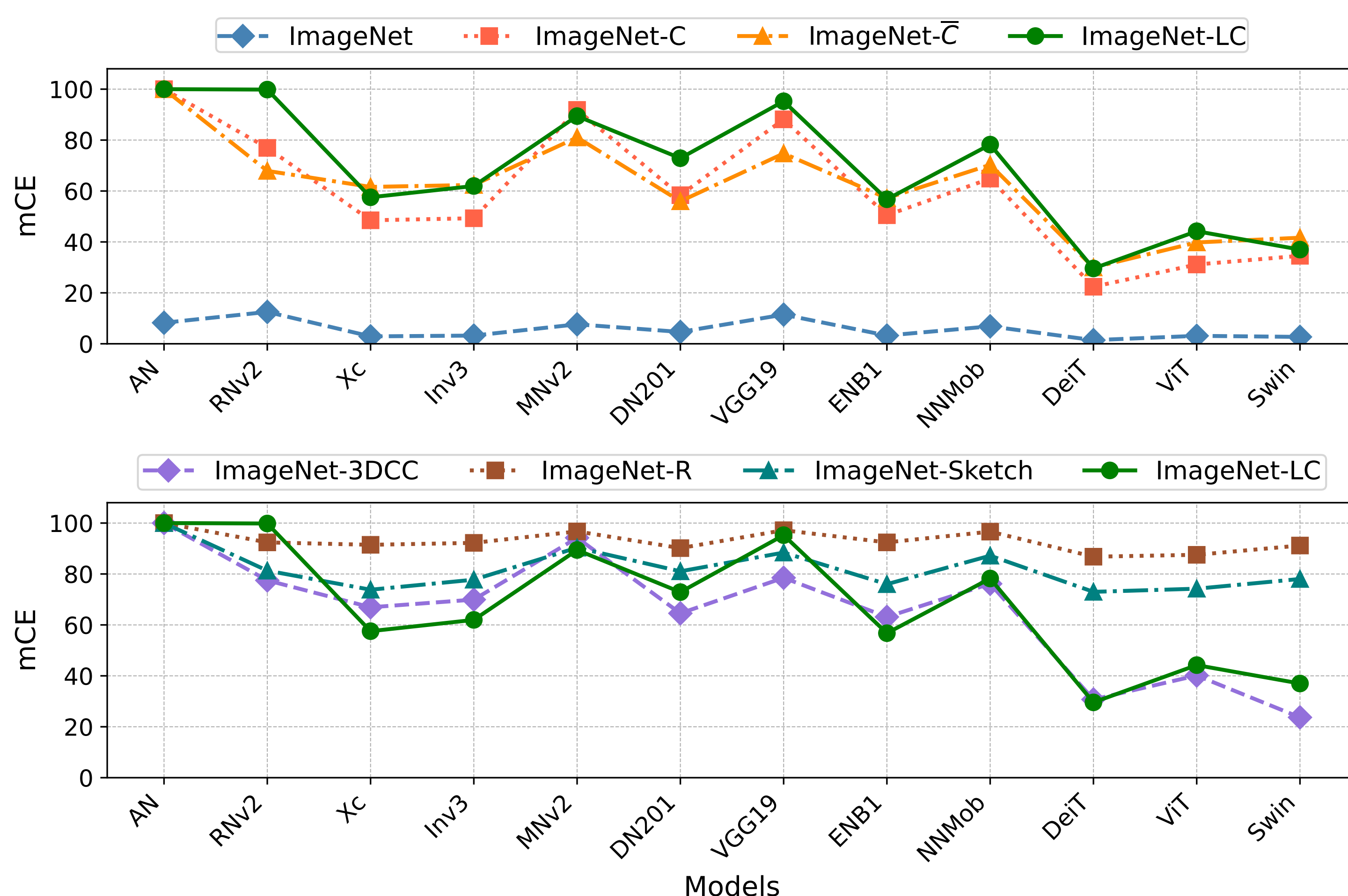


Figure 2: mCE on IMAGENET-LC, normalised to AlexNet.

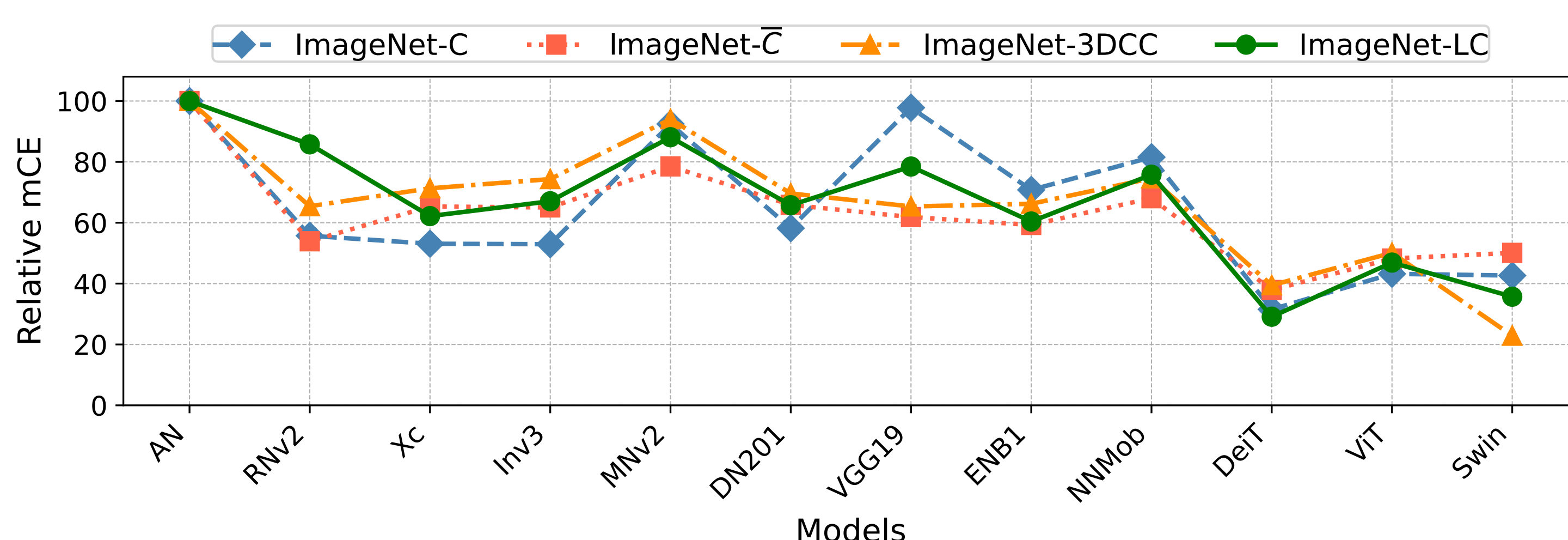


Figure 3: Relative mCE trend of IMAGENET-LC with other benchmarks.

Abbreviations for the above mentioned CNN and Transformer models: AN (AlexNet), RNv2 (ResNet50v2), Xc (Xception), Inv3 (Inceptionv3), MNv2 (MobileNetv2), DN201 (DenseNet201), ENB1 (EfficientNetB1), NNMob (NASNet-Mobile), and Swin (Swin Transformer)

Evaluation: Stealthiness

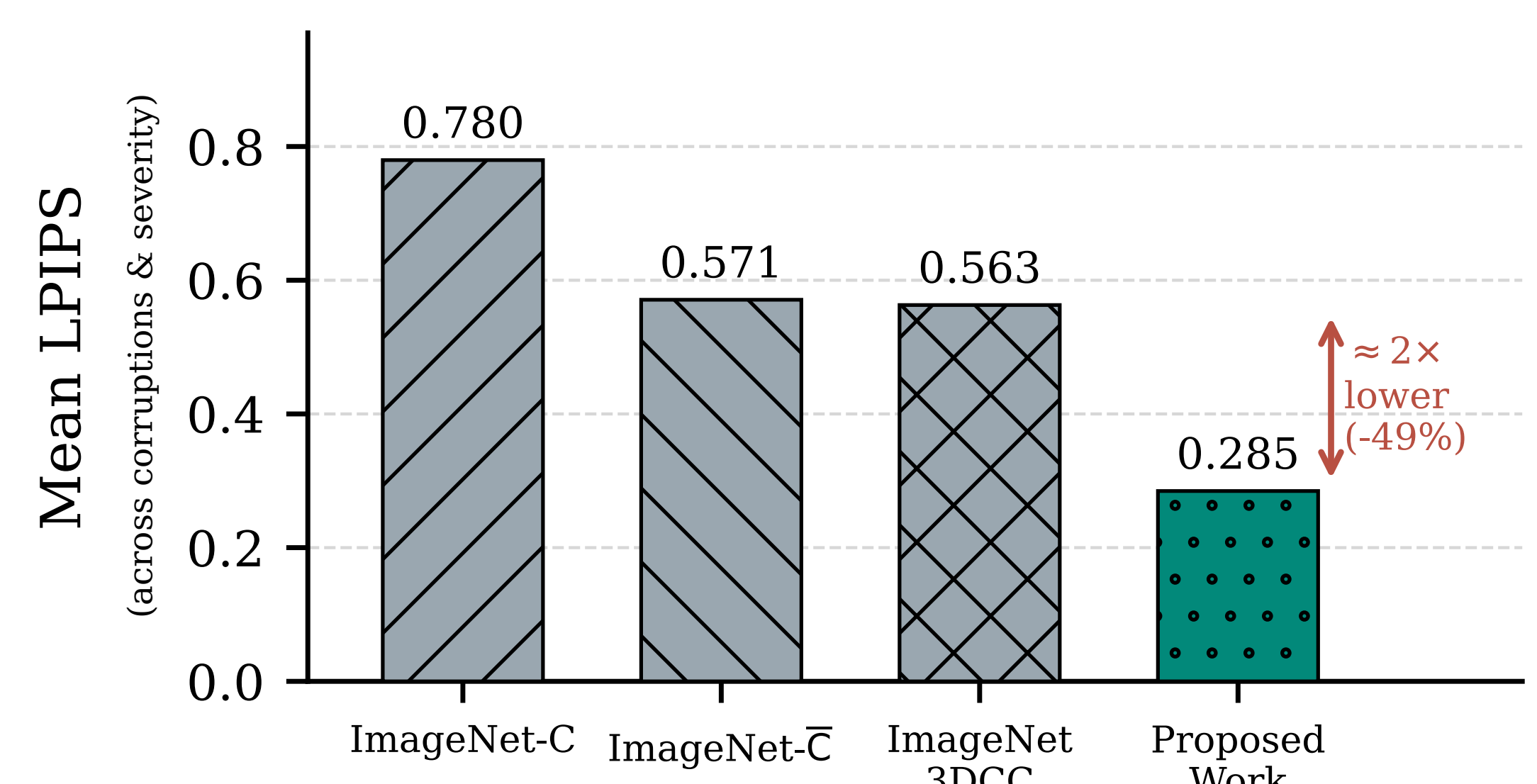


Figure 4: LPIPS score comparison across benchmarks.

- Every ViT beats every CNN; DeiT is the most robust model at mCE = 29.6.
- IMAGENET-LC sits half below every global benchmark across severity levels.
- ImageNet-LC benchmark exposes critical vulnerability of ML models under semantic natural corruptions.

Conclusion

IMAGENET-LC isolates object-centric, localized corruptions applied to semantic RoIs. Semantic preserving natural corruptions in the input space lowers the performance of ML model at least by 30% while being perceptually similar to the original input.

Future Work. Extend IMAGENET-LC to *dynamic and temporally evolving localized corruptions* for more realistic robustness evaluation.

References

- [1] Hendrycks & Dietterich. Benchmarking neural network robustness to common corruptions. *ICLR*, 2019.
- [2] Mintun *et al.* On interaction between augmentations and corruptions. *NeurIPS*, vol. 34, pp. 3571–3583, 2021.
- [3] Kar *et al.* 3D common corruptions and data augmentation. *CVPR*, pp. 18963–18974, 2022.
- [4] Swain *et al.* SPAT: semantic-preserving adversarial transformation. *ECAI*, vol. 372, pp. 2266–2273, 2023.